

1. **Batch processing** this is the data processing techniques where the larger volume of data are collected over a period of time and processed as a single unit at schedule interval.

Example of batch processing are: A retail company might use the batch processing to analyze customer purchase history from the previous month.

Why batch processing:

- Handle the larger volume of data
- Cost effective: It can reduce the operation cost compare to other system that require the continuous real time processing
- Resource Optimization.
ubores haji wa rasilimali

When to use Batch Processing:

- Historical data analysis.
 - Data transformation.
 - Generate larger scale report.
 - Periodic updates.
2. Real time processing: Refer to the ability of collecting, processing and analyze data, as it is being generated, with result available almost instantl.

It is categorized into two types namely;

- Transaction processing.
 - Stream processing: This is the data processing where data is continuously processed as It arrives. Eg A machine used to translate the language during international conference.
A live television channel.
3. What is the difference between the online data processing and stream line processing;

answer

Online Data Processing: Handles individual transactions in real time, focusing on quick response and consistency (e.g., banking systems).

Stream Data Processing: Processes continuous flows of data in real time, focusing on scalability and low-latency analytics (e.g., IoT data or social media streams).